

Education

York University

Bachelor of Science (Hons) in Computer Science GPA: 8.6 / 9

Fall 2023 - Summer 2027 (Expected)

Toronto, ON

Relevant Coursework: Object Oriented Programming (OOP), Data Structures, Design and Analysis of Algorithms, Software Tools and Scripting, Building E-Commerce Systems, Introduction to Database Systems, Data Mining, Technical Countermeasures to Financial Crimes, Intro to Computer Security

Work Experience

RBC

May 2026–Present

Data Analyst Co-op

Toronto, ON

- Built Tableau talent dashboards for executive-level reporting, translating complex workforce datasets into actionable insights that supported strategic decision-making within RBC's Operations Process Engineering team.
- Engineered Cleaned, validated, and transformed large-scale HR and operational datasets using Python and SQL, ensuring data integrity and consistency across workforce management pipelines and stakeholder-facing reports.
- Collaborated with cross-functional stakeholders to streamline operational processes, surfacing workforce insights and contributing to presentations that informed headcount planning and process improvement initiatives.

Sarit Micromobility, York University

Jan 2026–May 2026

Research Assistant

Toronto, ON

- Developed Python-based telemetry and data processing pipelines for next-generation micro-mobility vehicles, enabling real-time system monitoring and improving reliability across hardware and software layers.
- Engineered an RF and cloud-based remote lock/unlock system on Raspberry Pi, interfacing with electrical circuits and mechanical components to deliver secure, real-time vehicle access control for SARIT autonomous platforms.
- Collaborated within an interdisciplinary research team at York University to build, test, and document embedded software components, bridging electrical, mechanical, and software systems for safety-critical micro-mobility applications.

Elder Lab, York University

May 2025–Aug 2025

Researcher Assistant

Toronto, ON

- Conducted drone-based traffic analysis by collecting aerial video data and deploying YOLO models for multi-class vehicle detection; streamlined drone calibration and model tuning using agile project management, improving classification accuracy and reducing calibration time by **30%**.
- Utilized CVAT with Docker, Python (OpenCV, NumPy, Matplotlib, Pandas), ffmpeg, and Bash scripting for semi-automatic ground truthing, annotation, and preprocessing of large-scale image datasets, accelerating data labelling and preparation for deep learning workflows.
- Developed and maintained scalable computer vision pipelines for object detection and tracking in a Linux, UNIX, and hybrid cloud environment using Git, Bash, and CI/CD tools, improving deployment speed by **40%** and enabling reproducible workflows across **3+ active research projects with GPU (CUDA)** acceleration integrated into the rendering pipeline.

Projects

ConsultHub | Java, Spring Boot, PostgreSQL, Gemini API, Docker

January 2026

- Architected a full-stack consulting booking REST API using Spring Boot and PostgreSQL, implementing a 6-state booking lifecycle via the State design pattern, eliminating conditional transition logic across 15+ endpoints.
- Integrated a Gemini-powered AI customer assistant using **function calling (tool use)** to dynamically inject platform context such as services, availability, and policies into the model at runtime, delivering accurate recommendations without any model retraining or fine-tuning.
- Applied Strategy and Factory design patterns to build an extensible payment processing module supporting 4 payment methods (credit, debit, PayPal, bank transfer), containerized the full stack using Docker Compose with isolated PostgreSQL, H2 test, and Neon cloud database environments.

Fake News Classifier | Python, Pandas, NLP

September 2024

- Built an LSTM-based fake news classifier leveraging NLP and RNN techniques, boosting model accuracy to 91% through dropout regularization and hyperparameter tuning.
- Increased the model's accuracy to **91%** by adding a dropout layer and hyperparameter tuning. Cleaned **JSON/XML-based** data, pre-processed it using stemming, and applied **data science, statistics, and machine learning** for performance enhancement using the stemming technique, and vectorised it using the word embedding technique.
- Led 5+ peer review sessions to communicate model performance and architecture, demonstrating strong coaching and mentoring skills, and improving collaborative efficiency and code quality by 25%.

Movies Sentiment Analysis | Python, NLP, NumPy, Transformer

December 2023

- Applied Lemmatization, Tokenization, and TF-IDF to build a Bag of Words model for sentiment analysis on 5,000+ movie reviews, achieving 84% classification accuracy.
- Enhanced model performance by integrating BERT-based word embeddings, increasing accuracy to 87.5% and reducing false positives by approximately 15%.
- Utilized advanced analytics, parsed JSON/XML datasets, and developed visualizations to uncover sentiment patterns and ensure clear communication of model results, contributing to a 20% improvement in stakeholder interpretability.

Leadership and Volunteering

VP of Operations (AI and Data Science Society, York University) - Spearheaded logistics, event planning, and operational strategy for 15+ AI and data science initiatives, driving engagement and participation across 200+ students.

Peer Tutor (Excel Lassonde) - Provided academic tutoring and mentoring in computer science and math courses, helping 30+ peers improve understanding and boosting average academic performance.

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, Bash/Shell, Groovy, R, C++, HLSL, UML

Frameworks & Libraries: React, Angular, Vue.js, Node.js, Spring Boot, Django, SwiftUI, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Spark, Hive, Cypress, JUnit

Cloud & DevOps: AWS (Lambda, CloudFormation, CloudWatch), Azure, GCP, Docker, Kubernetes, Terraform, Ansible, Helm, Jenkins, CI/CD

Data & ML: Machine Learning, Deep Learning, NLP, ETL, Data Modelling, Big Data, Advanced Analytics, Snowplow, Databricks

Databases & BI: MySQL, PostgreSQL, Oracle, Snowflake/Data Warehouse, Tableau, Power BI, Grafana, Prometheus, Datadog

Tools & Platforms: Git, Bitbucket, Jira, Confluence, Jupyter, Nexus, Slack

Methodologies: Agile, Back-end Development, Hadoop